CLASS 10 · CBSE · NCERT MATHEMATICS · PYQ SOLUTIONS

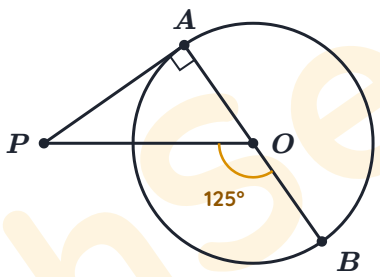
Chapter 10: Circles

Step-by-step solutions to the previous year questions

The question numbers match the PYQ question sheet.

1 Theorem 1: tangent and radius

1. PA is a tangent from an external point P to a circle with centre O, touching it at A. AOB is a diameter and $\angle POB = 125°$. Find $\angle APO$. (a) $25°$ (b) $65°$ (c) $90°$ (d) $35°$ CBSE 2026, 1 mark



SOLUTION

$\angle AOP = 180° - 125° = 55°$ (angles on the straight line AOB)

$\angle OAP = 90°$ (the radius is perpendicular to the tangent at the point of contact)

$\angle APO = 180° - 90° - 55° = 35°$

Answer: (d)

2. Two concentric circles have radii 5 cm and 4 cm. A chord of the larger circle touches the smaller circle. Find the length of the chord. CBSE 2026, 2 marks

SOLUTION

Let O be the centre and let the chord AB touch the smaller circle at M. Then $OM \perp AB$ and $OM = 4$ cm.

The perpendicular from the centre bisects the chord, so $AM = MB$.

In right triangle OMA: $AM = \sqrt{5^2 - 4^2} = 3$ cm

Answer: $AB = 6$ **cm**

3. Tangents are drawn to a circle at the two ends of a diameter. How are the two tangents related?

CBSE 2025, 1 mark

SOLUTION

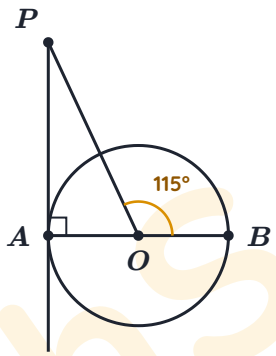
Each tangent is perpendicular to the radius at its point of contact, so both tangents are perpendicular to the same diameter.

Two lines perpendicular to the same line are parallel.

Answer: the tangents are parallel

4. AOB is a diameter of a circle with centre O, and PA is the tangent at A. If $\angle POB = 115°$, find $\angle APO$.

CBSE 2025, 1 mark



SOLUTION

$\angle AOP = 180° - 115° = 65°$ (angles on the straight line AOB)

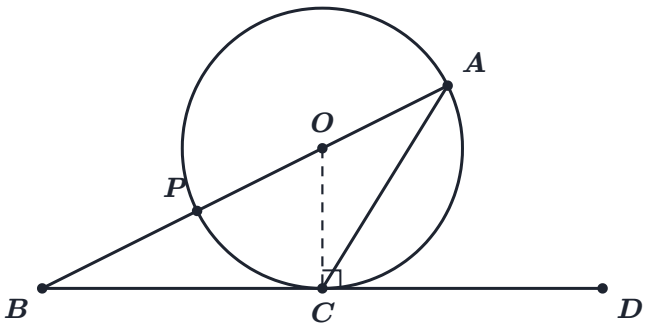
$\angle OAP = 90°$ (the tangent is perpendicular to the radius at the point of contact)

In $\triangle OAP$: $\angle APO = 180° - 90° - 65°$

Answer: $25°$

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5. O is the centre of a circle and BCD is the tangent to it at C. The line from B through O meets the circle at P and A. Prove that $\angle BAC + \angle ACD = 90°$. CBSE 2025, 2 marks



SOLUTION

$OA = OC$ (radii), so $\angle OAC = \angle OCA$ ... (1)

$OC \perp BD$, because the radius is perpendicular to the tangent at the point of contact. So $\angle OCA + \angle ACD = 90°$ ... (2)

A, O and B lie on one line, so $\angle BAC = \angle OAC$.

From (1) and (2): $\angle BAC + \angle ACD = 90°$. **Proved.**

6. Assertion (A): the tangents at the two ends of a diameter of a circle are parallel. Reason (R): the diameter is the longest chord of a circle. Choose: (a) A and R are both true and R explains A. (b) A and R are both true but R does not explain A. (c) A is true and R is false. (d) A is false and R is true.

CBSE 2024, 1 mark

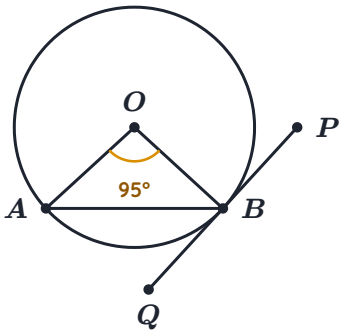
SOLUTION

A is true: each tangent is perpendicular to the diameter at its end point, and two lines perpendicular to the same line are parallel.

R is true, but it is not the reason why the tangents are parallel.

Answer: (b)

7. PQ is the tangent at B to a circle with centre O, and AB is a chord with $\angle AOB = 95°$. Q lies on the same side of OB as A. Find $\angle ABQ$. (a) $47.5°$ (b) $42.5°$ (c) $85°$ (d) $95°$ CBSE 2023, 1 mark



SOLUTION

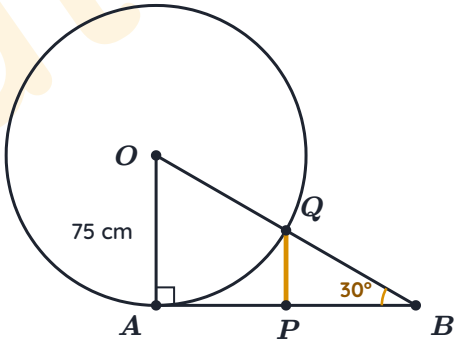
$OA = OB$ (radii), so $\angle OBA = \angle OAB = \dfrac{180° - 95°}{2} = 42.5°$.

$\angle OBQ = 90°$ (radius is perpendicular to tangent).

$\angle ABQ = 90° - 42.5° = 47.5°$

Answer: (a)

8. A discus leaves the thrower's circular path along a tangent. AB is such a tangent, touching a circle of radius 75 cm and centre O at A, with $\angle ABO = 30°$. OB cuts the circle at Q, and P is the point on AB with $PQ \parallel OA$. (i) Find AB. (ii) Find OB. (iii) Find AP, OR find PQ. CBSE 2023, 4 marks



SOLUTION

$OA \perp AB$, because the radius is perpendicular to the tangent at the point of contact.

(i) $\tan 30° = \dfrac{OA}{AB}$, so $AB = 75\sqrt{3}$ cm. $AB = 75\sqrt{3}$ **cm.**

(ii) $\sin 30° = \dfrac{OA}{OB}$, so $OB = 150$ cm. $OB = 150$ **cm.**

(iii) $OQ = 75$ cm (radius), so $QB = 150 - 75 = 75$ cm and Q is the mid-point of OB.

$PQ \parallel OA$, so $\triangle BPQ \sim \triangle BAO$ with $\dfrac{BP}{BA} = \dfrac{PQ}{AO} = \dfrac{BQ}{BO} = \dfrac{1}{2}$.

$AP = \dfrac{1}{2}AB$. $AP = \dfrac{75\sqrt{3}}{2}$ **cm.** OR: $PQ = \dfrac{1}{2}OA$. $PQ = 37.5$ **cm.**

9. Two concentric circles have radii 4 cm and 3 cm. Find the length of the chord of the larger circle that touches the smaller circle. CBSE 2022 Term 2, 2 marks

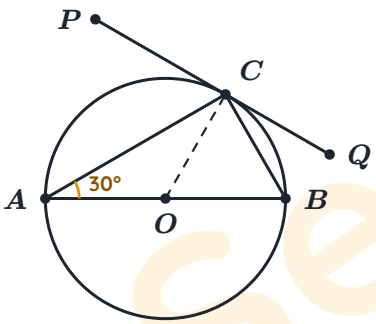
SOLUTION

Let O be the centre and let the chord AB touch the smaller circle at M. Then $OM \perp AB$, $OM = 3$ cm and $OA = 4$ cm.

$AM = \sqrt{4^2 - 3^2} = \sqrt{7}$ cm, and the perpendicular from the centre bisects the chord.

Answer: $AB = 2\sqrt{7}$ cm

10. PQ is the tangent at C to a circle with centre O, and AB is a diameter with $\angle CAB = 30°$. P lies on the same side of OC as A. Find $\angle PCA$. CBSE 2016, 1 mark



SOLUTION

$OA = OC$ (radii), so $\angle OCA = \angle OAC = 30°$.

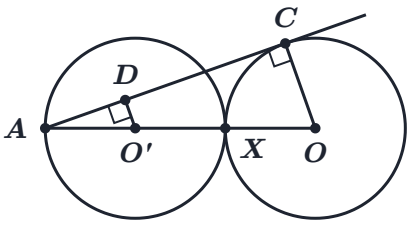
$\angle OCP = 90°$ (radius is perpendicular to tangent).

$\angle PCA = \angle OCP - \angle OCA = 90° - 30°$

Answer: $60°$

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11. Two equal circles with centres O and O' touch each other at X. The line OO' produced meets the circle with centre O' again at A. AC is a tangent from A to the circle with centre O, touching it at C, and $O'D \perp AC$. Find $\frac{DO'}{CO}$. CBSE 2016, 4 marks



SOLUTION

Let each radius be r. Then $AO' = r$ and $AO = AO' + O'X + XO = 3r$.

$\angle ACO = 90^\circ$ (radius is perpendicular to tangent) and $\angle ADO' = 90^\circ$ (given).

In $\triangle ADO'$ and $\triangle ACO$: $\angle A$ is common and $\angle ADO' = \angle ACO$, so $\triangle ADO' \sim \triangle ACO$ (AA).

$$\frac{DO'}{CO} = \frac{AO'}{AO} = \frac{r}{3r}$$

Answer: $\frac{1}{3}$

2 Theorem 2: tangents from an external point

12. TP and TQ are tangents from an external point T to a circle with centre O, and $\angle POQ = 120^\circ$. Find $\angle PTQ$. (a) 60° (b) 70° (c) 80° (d) 90° CBSE 2026, 1 mark

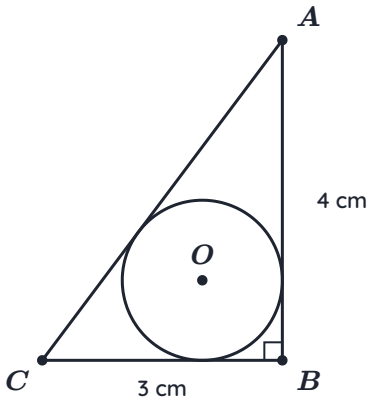
SOLUTION

$\angle OPT = \angle OQT = 90^\circ$ (radius is perpendicular to tangent).

In quadrilateral $OPTQ$: $\angle PTQ = 360^\circ - 90^\circ - 90^\circ - 120^\circ = 60^\circ$

Answer: (a)

13. $\triangle ABC$ is right-angled at B with $AB = 4$ cm and $BC = 3$ cm. Find the radius of the circle inscribed in the triangle. CBSE 2026, 3 marks



SOLUTION

$AC = \sqrt{4^2 + 3^2} = 5$ cm

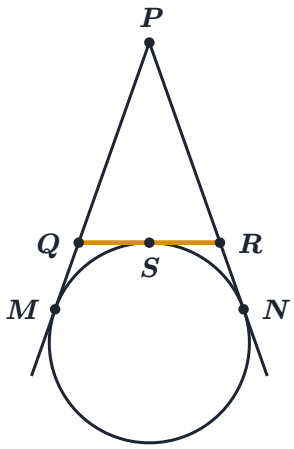
Let the radius be r and let the circle touch AB, BC and CA at D, E and F. $BDOE$ is a square, so $BD = BE = r$.

Tangents from a point are equal: $AF = AD = 4 - r$ and $CF = CE = 3 - r$.

$AC = AF + CF$: $5 = (4 - r) + (3 - r)$, so $2r = 2$

Answer: $r = 1$ **cm**

14. A circle touches the side QR of $\triangle PQR$ at S, and touches PQ and PR produced at M and N. Prove that $PM = \frac{1}{2}(PQ + QR + PR)$. CBSE 2026, 3 marks



SOLUTION

Tangents drawn from an external point to a circle are equal: $PM = PN$, $QM = QS$ and $RN = RS$.

$PQ + QR + PR = PQ + (QS + SR) + PR = (PQ + QM) + (PR + RN)$

$= PM + PN = 2PM$

So $PM = \frac{1}{2}(PQ + QR + PR)$**. Proved.**

15. A point P is 26 m from the centre of a circle. Each tangent from P to the circle is 10 m long. Find the radius of the circle. CBSE 2025, 2 marks

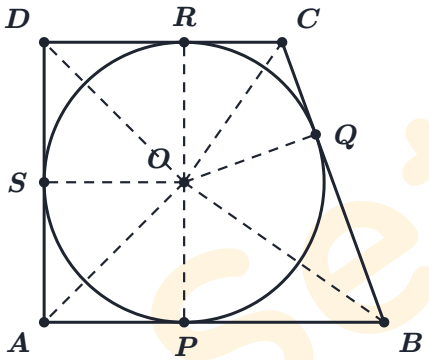
SOLUTION

Let O be the centre and T a point of contact. Then $\angle OTP = 90^\circ$.

$OT^2 = OP^2 - PT^2 = 26^2 - 10^2 = 676 - 100 = 576$

Answer: radius $= 24$ m

16. A quadrilateral is drawn around a circle so that all four sides touch it. Prove that each pair of opposite sides subtends supplementary angles at the centre. CBSE 2025, 2 marks



SOLUTION

Let $ABCD$ touch the circle, centre O, at P, Q, R and S on AB, BC, CD and DA.

In $\triangle OAP$ and $\triangle OAS$: $OP = OS$ (radii), $AP = AS$ (tangents from A) and OA is common. So the triangles are congruent (SSS) and $\angle AOP = \angle AOS = a$.

In the same way, $\angle BOP = \angle BOQ = b$, $\angle COQ = \angle COR = c$ and $\angle DOR = \angle DOS = d$.

The angles around O add up to 360°: $2a + 2b + 2c + 2d = 360^\circ$, so $a + b + c + d = 180^\circ$.

$\angle AOB = a + b$ and $\angle COD = c + d$, so $\angle AOB + \angle COD = 180^\circ$. Then $\angle BOC + \angle DOA = 360^\circ - 180^\circ = 180^\circ$ as well.

So opposite sides subtend supplementary angles at the centre. Proved.

17. AB is a diameter of a circle with centre O. The tangents at A and B are met by a third tangent at Q and P respectively. Prove that $\angle POQ = 90^\circ$. CBSE 2024, 3 marks

SOLUTION

Let PQ touch the circle at C. Join OC.

In $\triangle OAQ$ and $\triangle OCQ$: $OA = OC$ (radii), OQ is common and $QA = QC$ (tangents from Q). So $\triangle OAQ \cong \triangle OCQ$ (SSS), and $\angle AOQ = \angle COQ$.

Similarly $\triangle OBP \cong \triangle OCP$, so $\angle BOP = \angle COP$.

AOB is a straight line: $\angle AOQ + \angle COQ + \angle COP + \angle BOP = 180^\circ$, so $2\angle COQ + 2\angle COP = 180^\circ$.

Hence $\angle POQ = \angle COQ + \angle COP = 90^\circ$.

18. A circle of radius 8 cm with centre O touches the sides AD, DC, CB and BA of a quadrilateral $ABCD$ at P, Q, R and S. $AD \perp DC$, $BC = 30$ cm and $BS = 24$ cm. Find DC. CBSE 2024, 3 marks

SOLUTION

Tangents from B are equal: $BR = BS = 24$ cm, so $CR = BC - BR = 30 - 24 = 6$ cm.

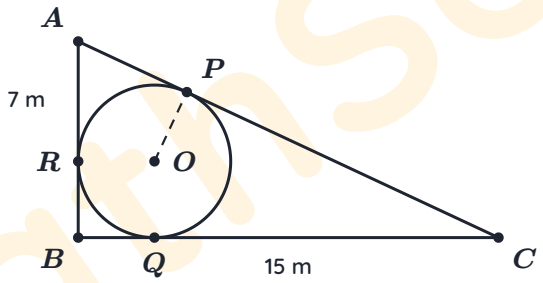
Tangents from C are equal: $CQ = CR = 6$ cm.

In quadrilateral $OPDQ$: $\angle D = 90^\circ$ (given) and $\angle OPD = \angle OQD = 90^\circ$ (radius is perpendicular to tangent), and $OP = OQ = 8$ cm. So $OPDQ$ is a square and $DQ = 8$ cm.

$DC = DQ + QC = 8 + 6$

Answer: $DC = 14$ **cm**

19. A triangular backyard ABC has a right angle at B, with $AB = 7$ m and $BC = 15$ m. A circular pit inside it touches AC, BC and AB at P, Q and R, and $AP = x$ m. (i) Write AR in terms of x. (ii) What kind of quadrilateral is $BQOR$? (iii) Write PC in terms of x and find x, OR find x and then the radius r of the pit. CBSE 2024, 4 marks, case study



SOLUTION

(i) Tangents from A are equal: $AR = AP = x$ m.

(ii) $\angle B = \angle OQB = \angle ORB = 90^\circ$ and $OQ = OR$ (radii), so $BQOR$ is a square.

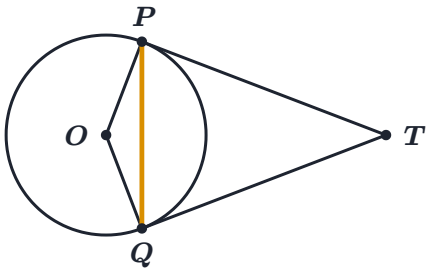
(iii) $BR = AB - AR = 7 - x$, so $BQ = 7 - x$ and $QC = 15 - (7 - x) = 8 + x$. Tangents from C are equal, so $PC = QC = (8 + x)$ m.

$AC = \sqrt{7^2 + 15^2} = \sqrt{274}$ m, and $AC = AP + PC = 2x + 8$. So $x = \dfrac{\sqrt{274} - 8}{2} \approx \dfrac{16.55 - 8}{2} \approx 4.28$.

OR: the radius equals the side of the square, $r = BR = 7 - x = \dfrac{22 - \sqrt{274}}{2} \approx 2.72$ m.

Answer: (i) x **m (ii) a square (iii)** $PC = (8 + x)$ **m,** $x \approx 4.28$**; or** $r \approx 2.72$ **m**

20. TP and TQ are tangents from an external point T to a circle with centre O. Prove that $\angle PTQ = 2\angle OPQ$. CBSE 2023, 3 marks



SOLUTION

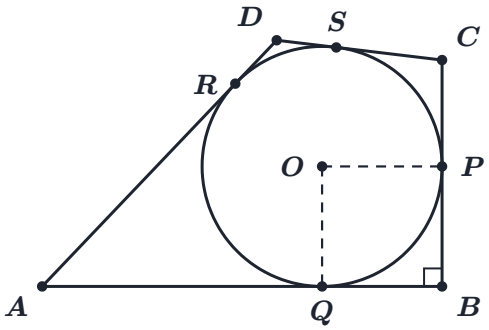
Let $\angle PTQ = \theta$. $TP = TQ$ (tangents from an external point), so $\triangle TPQ$ is isosceles.

$$\angle TPQ = \angle TQP = \frac{180^\circ - \theta}{2} = 90^\circ - \frac{\theta}{2}$$

$\angle OPT = 90^\circ$ (radius is perpendicular to tangent), so $\angle OPQ = 90^\circ - \left(90^\circ - \frac{\theta}{2}\right) = \frac{\theta}{2}$.

So $\angle PTQ = 2\angle OPQ$. Proved.

21. A circle with centre O is inscribed in a quadrilateral $ABCD$, touching AB, BC, CD and DA at Q, P, S and R. $\angle B = 90^\circ$, $AD = 17$ cm, $AB = 20$ cm and $DS = 3$ cm. Find the radius of the circle.
CBSE 2023, 3 marks



SOLUTION

Tangents from a point are equal: $DR = DS = 3$ cm, so $AR = AD - DR = 17 - 3 = 14$ cm.

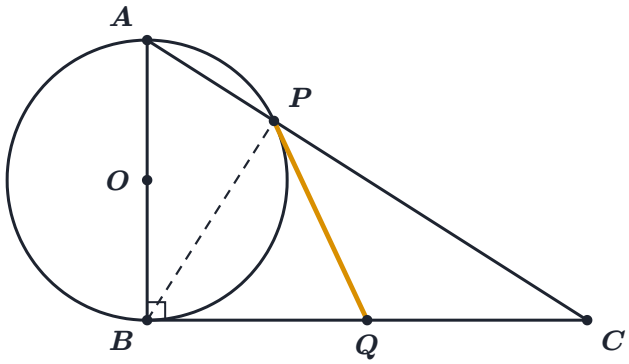
$AQ = AR = 14$ cm, so $QB = AB - AQ = 20 - 14 = 6$ cm.

In $OQBP$: $\angle B = 90^\circ$ and $\angle OQB = \angle OPB = 90^\circ$ (radius is perpendicular to tangent), and $OQ = OP$. So $OQBP$ is a square.

$r = OQ = QB$

Answer: radius $= 6$ cm

22. $\triangle ABC$ is right-angled at B. A circle drawn on AB as diameter cuts AC at P. The tangent to the circle at P meets BC at Q. Prove that Q is the mid-point of BC. CBSE 2022 Term 2, 4 marks



SOLUTION

AB is a diameter and $BC \perp AB$ at B, so BC is the tangent to the circle at B.

QB and QP are tangents from Q, so $QB = QP$ and $\angle QBP = \angle QPB$ ... (1)

$\angle APB = 90^\circ$ (angle in a semicircle), so $\angle BPC = 90^\circ$.

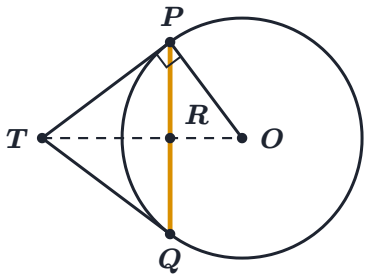
$\angle QPC = 90^\circ - \angle QPB$. In right triangle BPC, $\angle QCP = 90^\circ - \angle QBP$.

By (1), $\angle QPC = \angle QCP$, so $QP = QC$.

Hence $QB = QP = QC$. **So the tangent at P bisects BC. Proved.**

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23. PQ is a chord of length 8 cm in a circle of radius 5 cm with centre O. The tangents at P and Q meet at T. Find TP. CBSE 2019, 3 marks



SOLUTION

Join OT, meeting PQ at R. $TP = TQ$ and $OP = OQ$, so OT is the perpendicular bisector of PQ. Hence $PR = 4$ cm and $\angle PRO = 90^\circ$.

In $\triangle PRO$: $OR = \sqrt{5^2 - 4^2} = 3$ cm.

$\angle OPT = 90^\circ$ (radius is perpendicular to tangent). $\triangle TPO$ and $\triangle PRO$ both have a right angle and share $\angle POT$, so they are similar (AA).

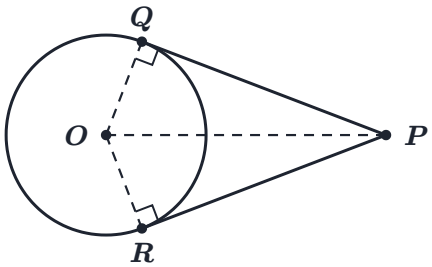
$\frac{TP}{PR} = \frac{PO}{RO}$, so $TP = \frac{4 \times 5}{3}$

Answer: $TP = \frac{20}{3}$ **cm, about 6.67 cm**

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24. Prove that the two tangents drawn to a circle from a point outside it are equal in length.

CBSE 2018, 3 marks



SOLUTION

Given: a circle with centre O, and tangents PQ and PR from an external point P. **To prove:** $PQ = PR$.

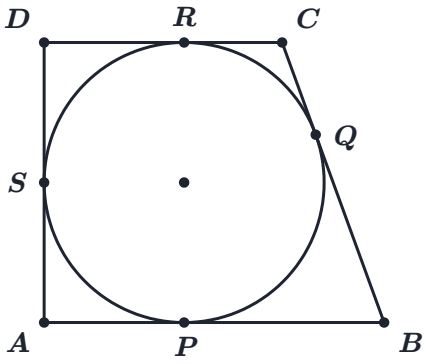
Join OP, OQ and OR. The radius is perpendicular to the tangent at the point of contact, so $\angle OQP = \angle ORP = 90°$.

In right triangles OQP and ORP: $OQ = OR$ (radii) and OP is common.

So $\triangle OQP \cong \triangle ORP$ by the RHS rule.

Hence $PQ = PR$. Proved.

25. A quadrilateral $ABCD$ is drawn around a circle, with AB, BC, CD and DA touching the circle at P, Q, R and S. Prove that $AB + CD = BC + DA$. CBSE 2016, 2 marks



SOLUTION

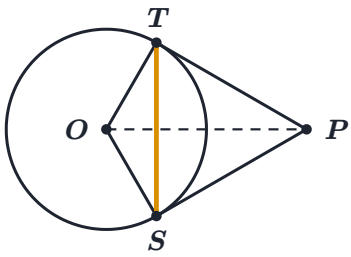
Tangents from an external point are equal: $AP = AS$, $BP = BQ$, $CR = CQ$ and $DR = DS$.

$AB + CD = (AP + BP) + (CR + DR) = AS + BQ + CQ + DS$

$= (AS + DS) + (BQ + CQ) = DA + BC$

So $AB + CD = BC + DA$. Proved.

26. PT and PS are tangents from an external point P to a circle with centre O and radius r. If $OP = 2r$, show that $\angle OTS = \angle OST = 30°$. CBSE 2016, 2 marks



SOLUTION

$\angle OTP = 90°$ (radius is perpendicular to tangent). In $\triangle OTP$: $\cos \angle TOP = \dfrac{OT}{OP} = \dfrac{r}{2r} = \dfrac{1}{2}$, so $\angle TOP = 60°$.

In the same way $\angle SOP = 60°$, so $\angle TOS = 120°$.

$OT = OS$ (radii), so $\angle OTS = \angle OST = \dfrac{180° - 120°}{2}$.

So $\angle OTS = \angle OST = 30°$. Proved.

27. Prove that the tangents drawn to a circle from an external point are equal in length. CBSE 2016, 4 marks

SOLUTION

Let PQ and PR be the tangents from P to a circle with centre O. Join OP, OQ and OR.

$\angle OQP = \angle ORP = 90°$ (radius is perpendicular to tangent), $OQ = OR$ (radii) and OP is common.

So $\triangle OQP \cong \triangle ORP$ (RHS), and $PQ = PR$. **Proved.** (The figure is with the 2018 question above.)

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